



EHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO
DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 06



To design and analyze the behavior of CMOS Logic Gates

Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	CALCULATION AND CODING	OBSERVATION/ RESULTS	SCORE

Performance:

Objective: To be familiar with the operation of the CMOS logic gates

Apparatus:

- P-MOS and N-MOS transistors
- Resistors
- Multimeter
- Connecting wire
- LED

Theoretical Background:

N-MOSFET Operation:

In order for drain current flow from drain to source in an N-channel MOSFET, the gate to source voltage must be greater than the specific device voltage(V_T), and the drain to source voltage must be greater than zero. The threshold voltage is dependent upon the physical dimensions and parameters for the MOSFET device. For enhancement only N-MOSFET transistor the threshold voltage is positive, and the enhancement depletion, the threshold voltage is negative.

Regions of Operations:

- **Cutoff:** $V_{gs} < V_T$

$$I_D(\text{OFF}) = 0$$

- **Linear:** $V_{gs} > V_T$ and $V_{gs} - V_T > V_{DS}$

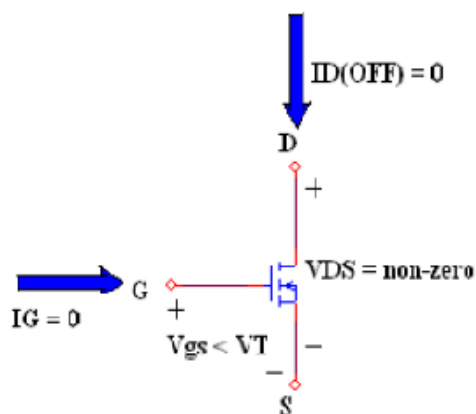
$$I_D(\text{linear}) = k(V_{gs} - V_T)V_{DS} - (V_{DS}^2)^2]$$

- **Saturation:** $V_{gs} > V_T$, $V_{DS} > V_{gs} - V_T$

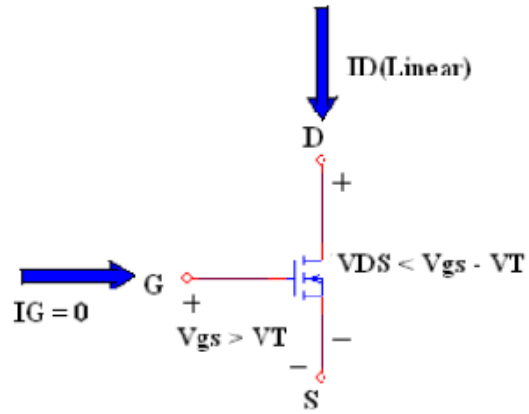
$$I_D(\text{sat}) = (k/2)*(V_{gs}-V_T)^2$$

Or

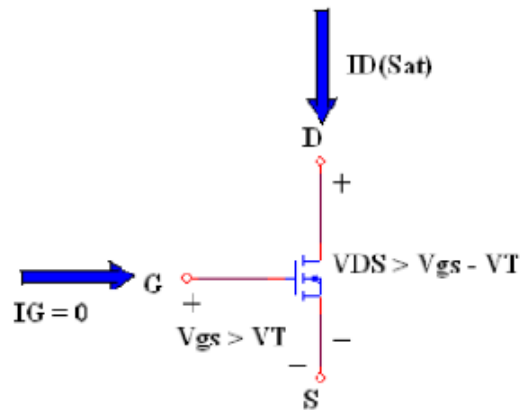
$$I_D(\text{sat}) = (k/2)*(V_{gs} - V_T)^2(1 + AV_{DS})$$



(a) Cutoff mode



(b) Linear mode



(c) Saturation mode

Figure-4.1(a, b, c) Shows the cutoff, linear and saturation mode

1. NMOS NOR GATE:

The general NMOS inverter can be augmented to perform the logical NOR function by placing additional output NMOS transistors in parallel with the output N-channel MOSFET. Figure-4.4

shows two input NOR gate with a generic load. But both NMOS transistors have their drain-to-source channels connected from the output to ground.

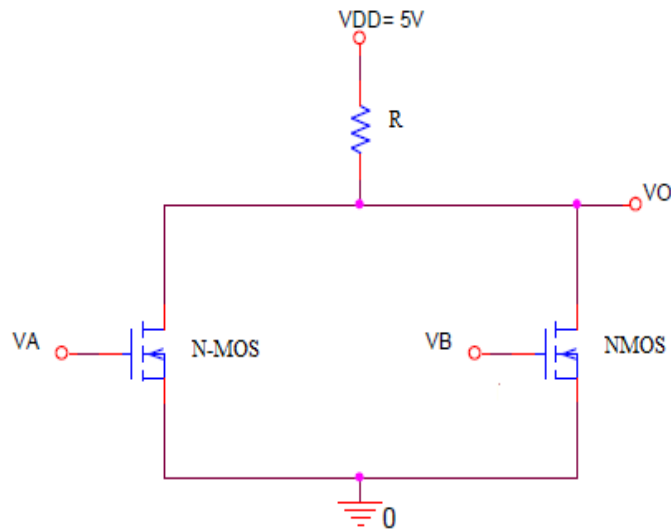


Figure-4.4 two input NOR gate using NMOS transistors

2. NMOS NAND Gate

NAND gates can also be easily constructed using NMOS circuitry Figure-4.5 shows a two input NMOS NOR gate with generic load.

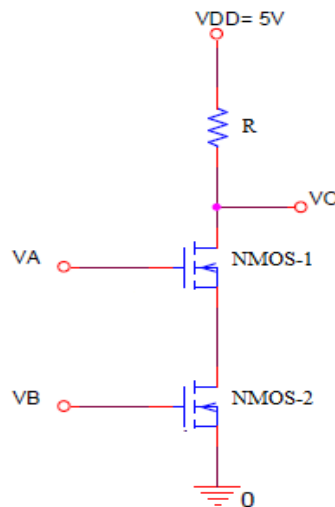


Figure-4.5 two input NMOS NAND Gate

3. CMOS OR and AND Gate

OR and NAND gates can also be easily constructed only adding the inverter circuit at the output of both logic circuits Figure-4.6 shows a two input OR gate and Figure-4.7 shows two input AND gate.

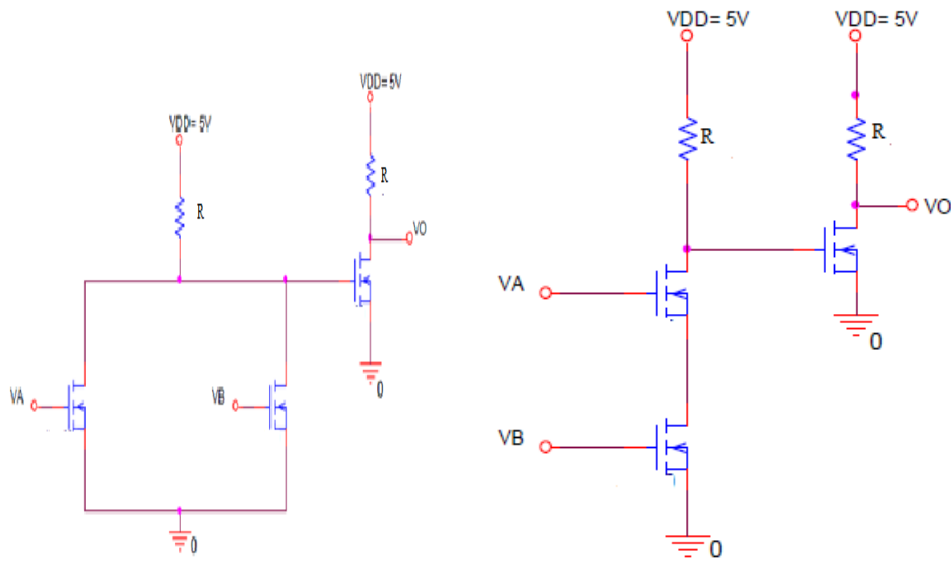


Figure-4.6 two input OR Gate Figure-4.7 two input AND Gate

Questions:

1. Why NMOS transistor is selected as pull-down transistor?

A pull-down transistor should bring output to “0” means it should provide a path to the charges to go into ground terminal. NMOS transistor is selected as pull-down transistor because it provides path to ground without any threshold voltage drop i.e. V_T . It is good for LOW pass logic “0”.

2. Why PMOS is connected to VDD?

In these devices, gate is reverse biased. PMOS consist of p-channel and to reverse bias the gate, a negative voltage is applied between gate and source. Therefore, PMOS (source terminal) is connected to VDD.

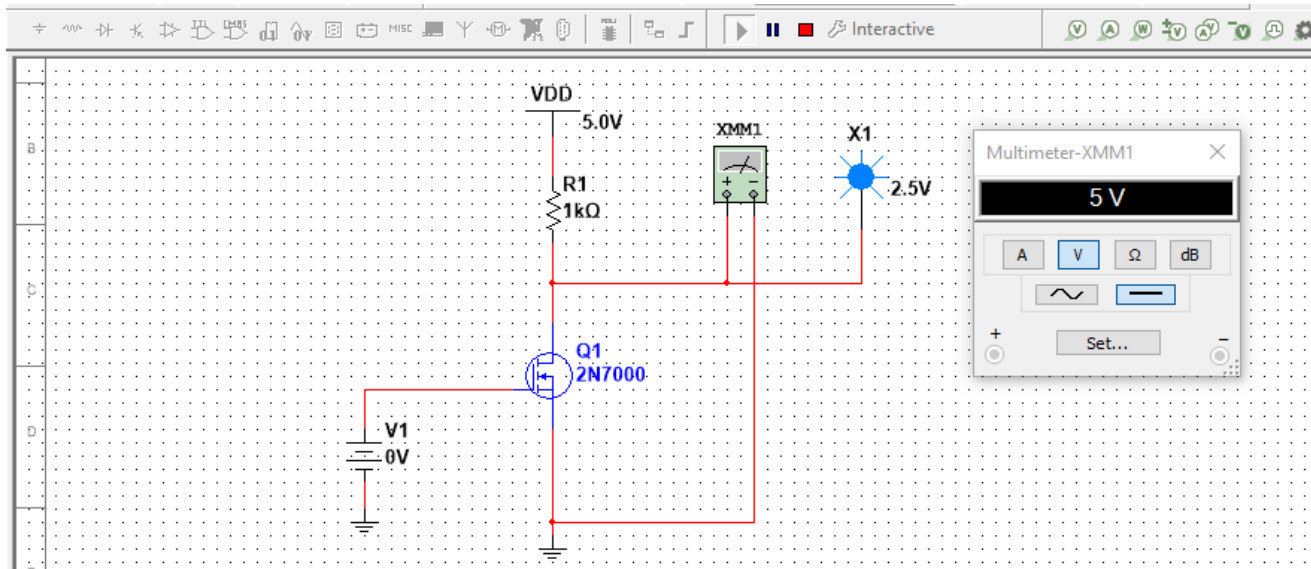
3. Which is better PMOS OR NMOS?

NMOS is better than PMOS because it has advantage of speed over PMOS because mobility of carriers (electrons) is greater than that of holes in PMOS.

Lab Task-1: NMOS Inverter Gate

- Connect the circuit shown in Figure-4.2, $V_{DD} = 5V$ and value of resistor by your choice.
- Complete the following table.

V_{IN}	V_{OUT}
0	5V
5	220.189mV



- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of Inverter logic gate.

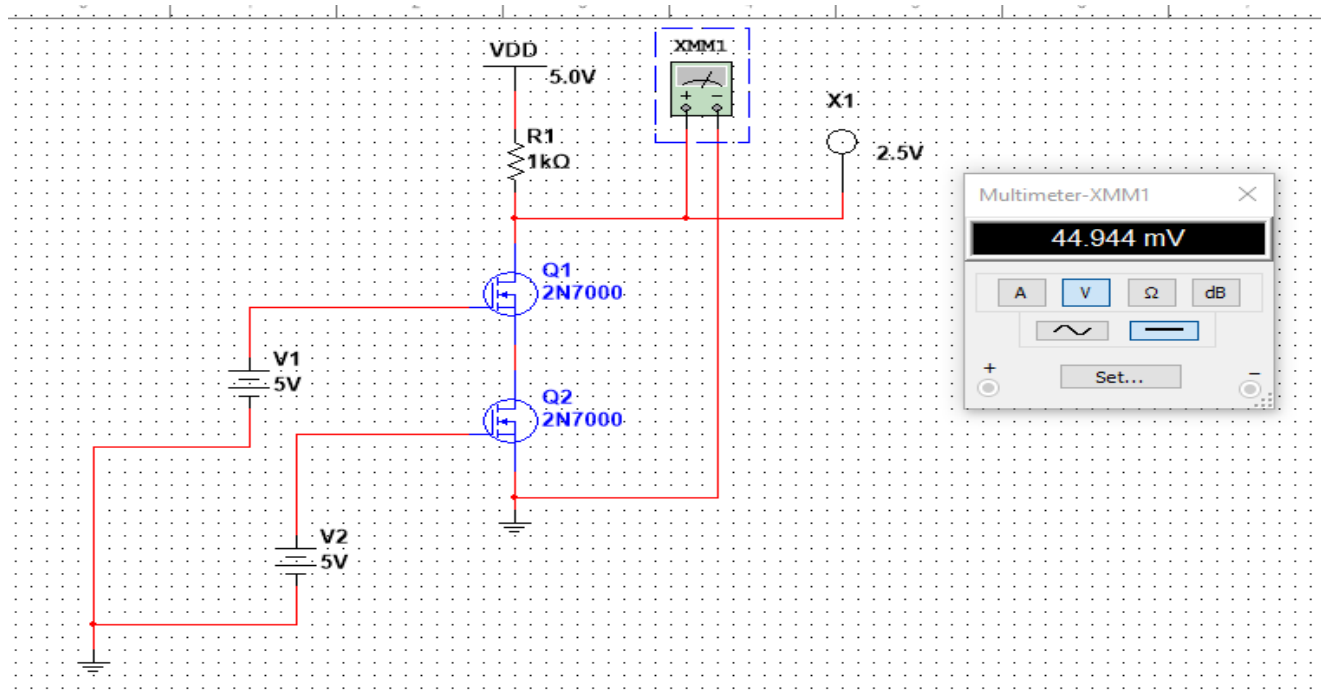
Vin	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
Vout	5V	5V	5V	5V	5V	3.741V	676.676mV	399.45mV	305mV	253.4mV	231.19mV

- Determine V_{OH} , V_{OL} , V_{IH} , V_{IL}

Lab Task-2: NMOS NAND Gate

- Connect the circuit shown in Figure-4.5, $V_{DD} = 5V$ and value of resistor by your choice. Complete the following table.

VA	VB	VOUT
0	0	5V
0	5	5V
5	0	5V
5	5	47.94mV



- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of NAND logic gate.

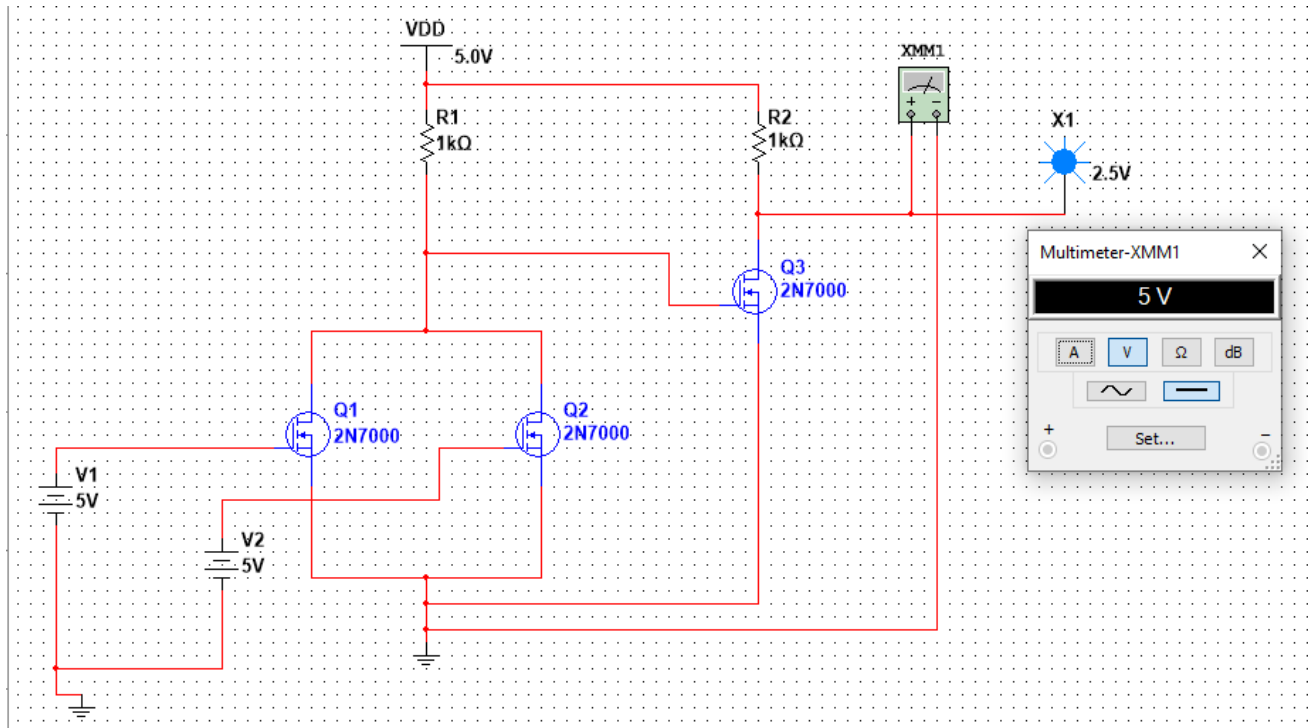
Vin	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
Vout	5V	5V	5V	5V	5V	265.76mV	114.316mV	77.6mV	61.6mV	55.58mV	44.94mV

- Determine V_{OH} , V_{OL} , V_{IH} , V_{IL}

Lab Task-3: NMOS OR Gate

- Connect the circuit shown in Figure-4.6, $V_{DD} = 5V$ and value of resistor by your choice.
- Complete the following table.

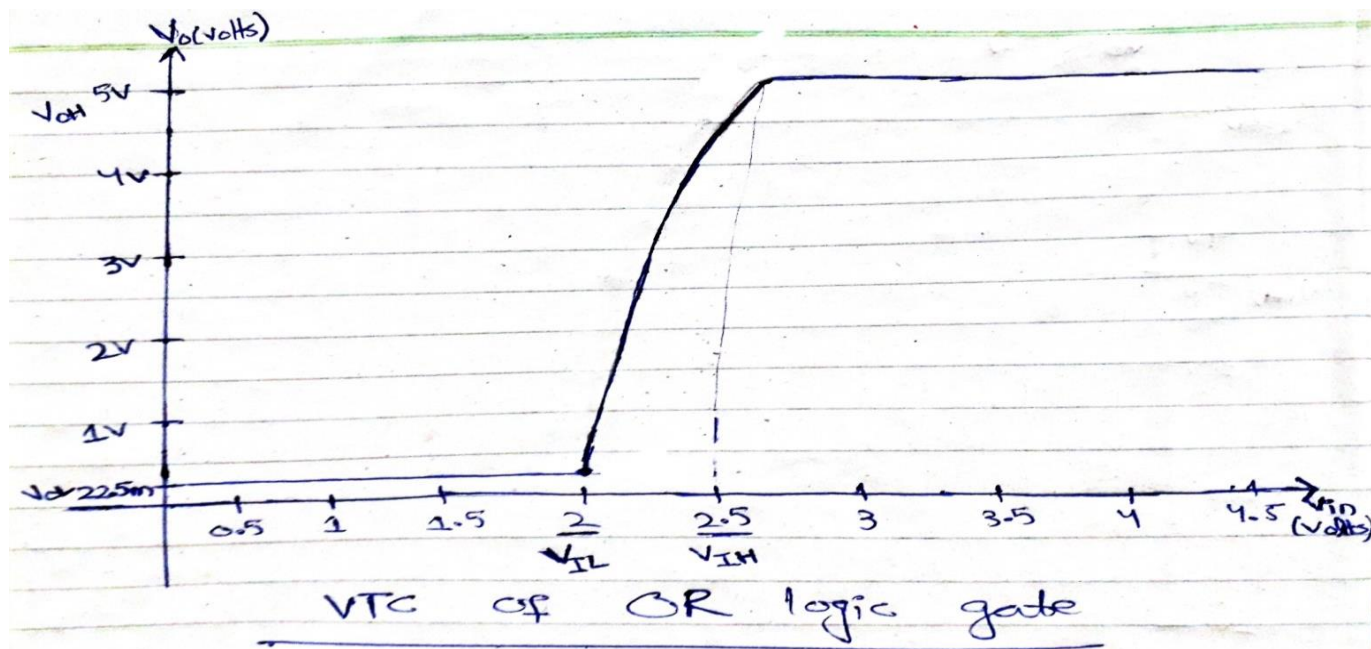
VA	VB	VOUT
0	0	22.5mV
0	5	5V
5	0	5V
5	5	5V



- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of OR logic gate.

V _{in}	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
V _{out}	22.5mV	22.5mV	22.5mV	22.5mV	22.5mV	5V	5V	5V	5V	5V	5V

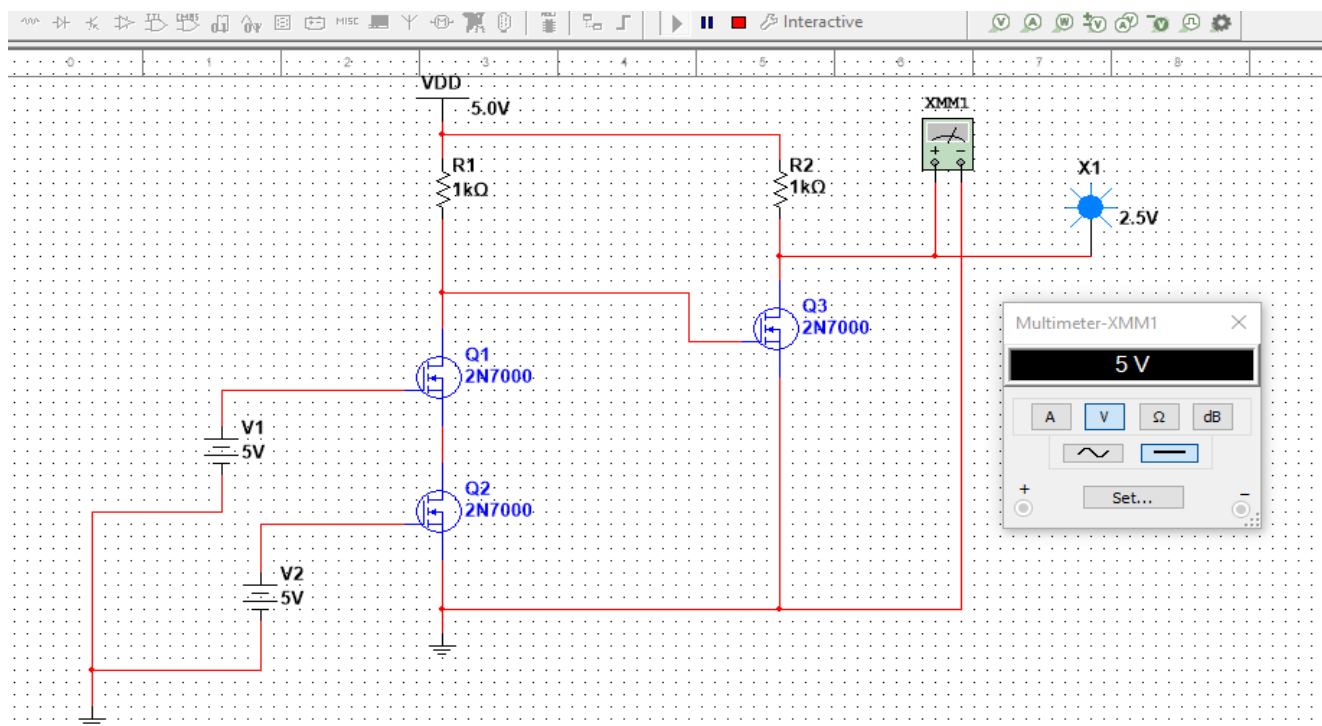
- Determine V_{OH} , V_{OL} , V_{IH} , V_{IL}



Lab Task-4: NMOS AND Gate

- Connect the circuit shown in Figure-4.7, $V_{DD} = 5V$ and value of resistor by your choice. Complete the following table.

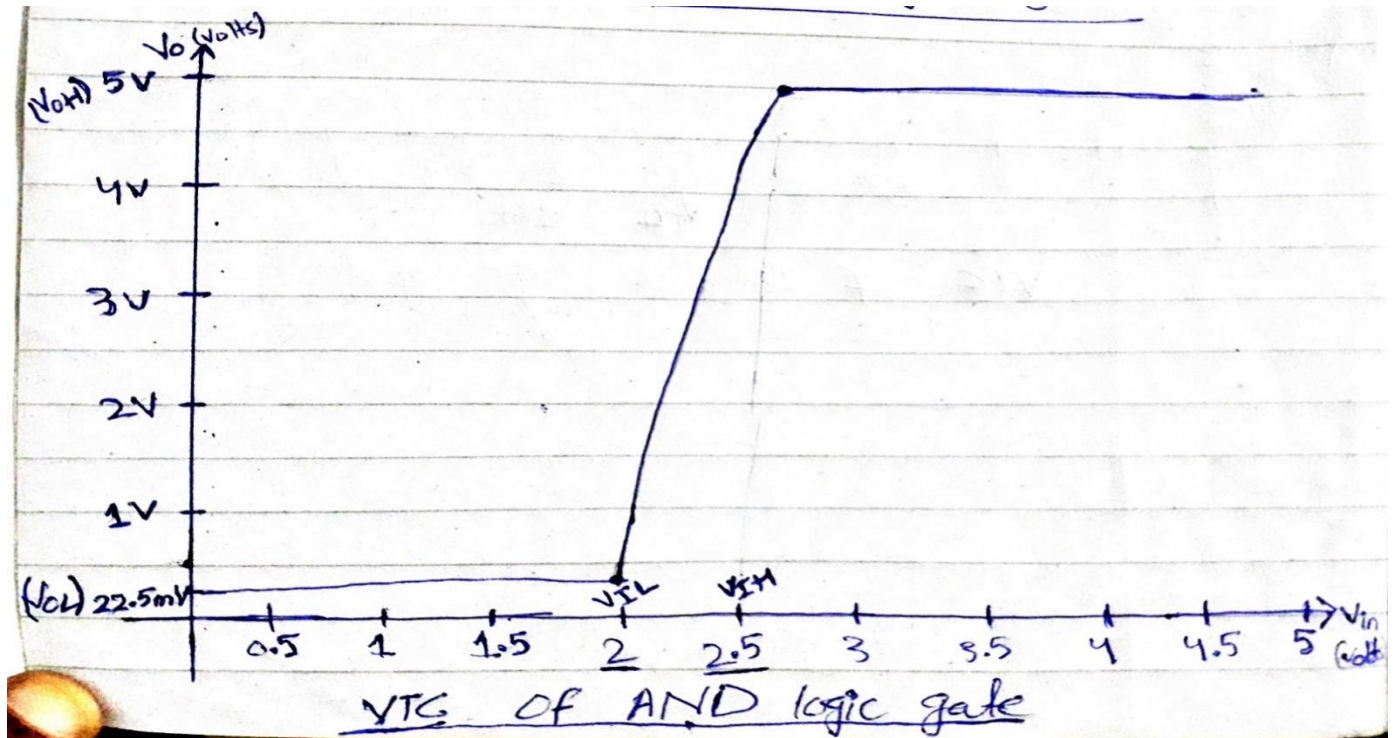
VA	VB	VOUT
0	0	22.5mV
0	5	22.5mV
5	0	22.5mV
5	5	5V



- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of AND logic gate.

Vin	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
Vout											

- Determine V_{OH} , V_{OL} , V_{IH} , V_{IL}



Conclusion: